

Comparison of dimension reduction techniques applied to the analysis of airborne radionuclide activity concentration

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ABSTRACT

Temporal variation of airborne radionuclide activity concentration is highly influenced by several meteorological parameters. A traditional key tool to perform a combined analysis on these data is Principal Component Analysis (PCA), a linear dimensionality reduction technique that prioritizes the conservation of the data set global structure. While it reveals important information regarding the correlation among the considered variables, the obtained visual representations do not usually allow to clearly discern different clusters of states with common properties. The main goal of this study is applying two recently introduced non linear dimensionality reduction techniques, t-distributed Stochastic Neighbor Embedding (t-SNE) and Uniform Manifold Approximation and Projection (UMAP) to a data set composed by ^7Be and gross beta (A_{β}) activity concentration and other meteorological data gathered in Mallorca (Spain) between 2004 and 2014. Compared to PCA, both algorithms reveal more details on the local structure of the data set. UMAP allows to clearly identify data clusters with different characteristics that are not clearly identified with the alternative techniques.

1. Introduction

The existence of radioactivity in the atmosphere is mainly due to the existence of cosmogenic and naturally occurring radionuclides, although anthropogenic radionuclides released in radioactive accidents and nuclear bomb testing can also be found (Atwood, 2013). Regular sampling and determination of airborne radionuclides is then a key part in most surveillance programmes with the main goal to establish background levels, and support decision making in emergency cases, as well as to use such radio tracers to model aerosol transport in the atmosphere (Gaffney et al., 2004; Papastefanou, 2008; Sangiorgi et al., 2019).

Airborne radioactivity has been extensively studied in connection to meteorological parameters that affect its temporal variation, focusing on the existing correlations between them (Kitto et al., 2005; Dueñas et al., 2009; Pham et al., 2011; Tositti et al., 2014; Gordo et al., 2015; Ajtić et al., 2018) as well as identifying clusters with similar properties (Piñero-García and Ferro-García, 2013). Several statistical tools have also been applied to forecast airborne radioactivity levels based on previous values of the time series (Dueñas et al., 2001; Bas et al., 2017b) as well as from the actual value of their correlated meteorological variables (Dueñas et al., 1999; Bas et al., 2017a; Chham et al., 2018).

^7Be is a cosmogenic radionuclide produced by spallation of cosmic

radiation with light atmospheric nuclei. It is produced mainly in the lower stratosphere and upper troposphere. Newly formed ^7Be quickly associates with small aerosols ($< 1 \mu\text{m}$ diameter) and is transported following the movement of the air mass in which it is located. Stratospheric ^7Be enters the troposphere with the stratospheric intrusions of aerosols into the troposphere, while wet and dry depositional events allow ^7Be to reach the surface. ^7Be has then been used to study and characterize air masses properties and transport through the troposphere (Todorovic et al., 2005; Leppänen et al., 2010; Bourcier et al., 2011; Piñero-García et al., 2015; Alegría et al., 2020).

Gross Beta measurements can be used to screen and detect concerning levels of radioactivity in the atmosphere, as well as to obtain information on aerosol residence times (Hernández et al., 2005).

The best way to understand the role of the considered variables on the performed measurements is to represent and visualize them, although this task is hard to accomplish in a high dimensional space. Principal Component Analysis (PCA) reduces the dimensionality of the system allowing to represent the measurements in a two or three dimensional plot, while losing as little information as possible in respect of the original data set. This technique is commonly used to visualize ^7Be activity concentrations and relate them to meteorological parameters (Piñero-García and Ferro-García, 2013; Gordo et al., 2015; San Miguel

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et al., 2019) helping to understand the role of the local climate. Its main drawback is that clustering identification is hard as all points tend to be quite mixed in the obtained lower dimensional space. Quite recently, alternative dimension reduction techniques like t-SNE and UMAP have appeared, providing a better local structure preservation in the performed transformation to reduce the dimensionality of the system (van der Maaten and Hinton, 2008; van der Maaten, 2014; McInnes et al., 2018, 2020). These techniques have recently been applied to a wide range of fields being especially successful in molecular biology studies devoted to the interpretation of single cell data analysis (Becht et al., 2019; Cao et al., 2019; Kobak and Berens, 2019).

The main aim of this work is to apply these techniques to a data set of airborne radionuclide activities and other correlated meteorological parameters, compare the visualizations obtained from them, and establish their strong and weak points. After tuning the model parameters, these techniques share the ability to preserve the local structure of the data set and allow to reveal the existence of well defined separated clusters with different properties.

2. Material and methods

2.1. Study area and surface air sampling

Airborne particulate samples were collected at the University of the Balearic Islands campus (39°38'17" N, 2°38'47" E) on a weekly basis, Fig. 1. The sampling point is located on Mallorca island, situated in the western Mediterranean Sea. Its Mediterranean climate is characterized by relatively mild winters and hot summer seasons.

Aerosol samples were collected using a Radeco AVS-28A low-volume sampler. The samples were collected using nitrocellulose filters of 4.7 cm diameter and 0.8 μm pore size, with a flow rate of 1.8 m^3h^{-1} . Each filter was changed weekly resulting in a total collected volume of approximately 300 m^3 per filter.

2.2. Gross Beta and ^7Be

Filters were placed on a stainless steel planchet, and gross beta (A_{β}) activity was measured with a Tennelec LB 4100 gas flow low background proportional counter for 1000 min. Filters were kept in the desiccator for six days prior to measurement to ensure the decay of all short lived radionuclides.

The detector was calibrated for efficiency, self-absorption and spill-over with a $^{90}\text{Sr}/^{90}\text{Y}$ standard obtained from CIEMAT. Background was checked monthly and an average value of 0.7 cpm was obtained.

^7Be measurement was performed with a high-purity Germanium detector (Canberra GR5023-7500 SL) through its 477 keV gamma emission. The detector was calibrated with a multi radionuclide standard from CIEMAT in the energy range of (46.2 keV–1467 keV). Given the typical low activity of ^7Be in the weekly gathered filters, the activity for 4 or 5 filters is measured in a combined monthly measure.

2.3. Meteorological parameters

Particulate Matter (PM) in the filters was determined by difference of masses, weighting the filter with a high precision analytical balance before and after the sampling was performed.

The rest of the atmospheric factors selected are precipitation (Prec), temperature (Temp), wind speed (WS) and relative humidity (RH). This selection has been performed considering previous studies focusing on the relation between the studied radionuclides and the meteorological variables usually correlated to them in other locations with a similar Mediterranean climate and medium latitude locations (Pham et al., 2011; Tositti et al., 2014; Gordo et al., 2015; Rodas Ceballos et al., 2016; Sáez-Muñoz et al., 2018). All these values were obtained from the Spanish Meteorological Agency (AEMET) station network.

2.4. Data set characterization

Temporal evolution of the variables included in the data set is represented in Fig. 2, where it can be observed that all of them follow a seasonal pattern. This seasonality is deeply relevant for the ^7Be , gross beta activity concentrations, dust content, and temperature, with maximum values during the summer season and minimum values in winter. The relative humidity also follows a seasonal pattern although the maximum values are reached during winter and the minimum values correspond to the summer season. The seasonal behavior is weaker for the precipitation and the wind speed, although it is noticeable that the most intense rainy seasons usually appear at the end of the summer and during autumn. The behavior and characteristic values for all data set variables, which are summarized in Table 1, are characteristic of the western Mediterranean Sea climate. The seasonal maximum and minimum values for the ^7Be and gross beta activity concentrations have

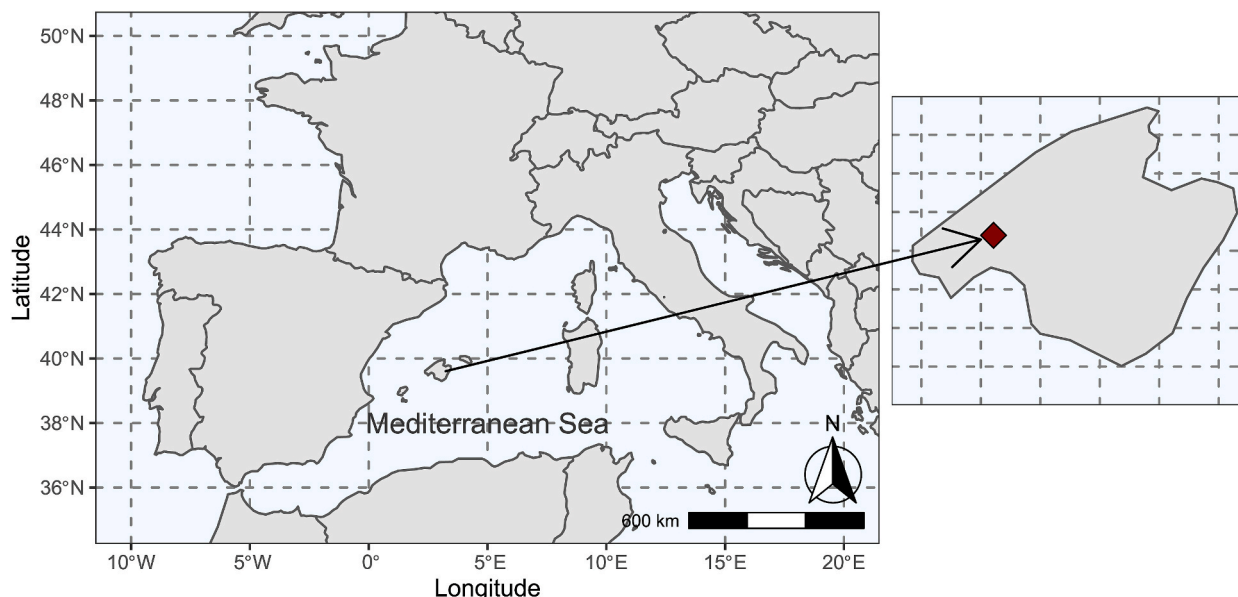


Fig. 1. Sampling location at the University of the Balearic Islands campus (39°38'17" N, 2°38'47" E).

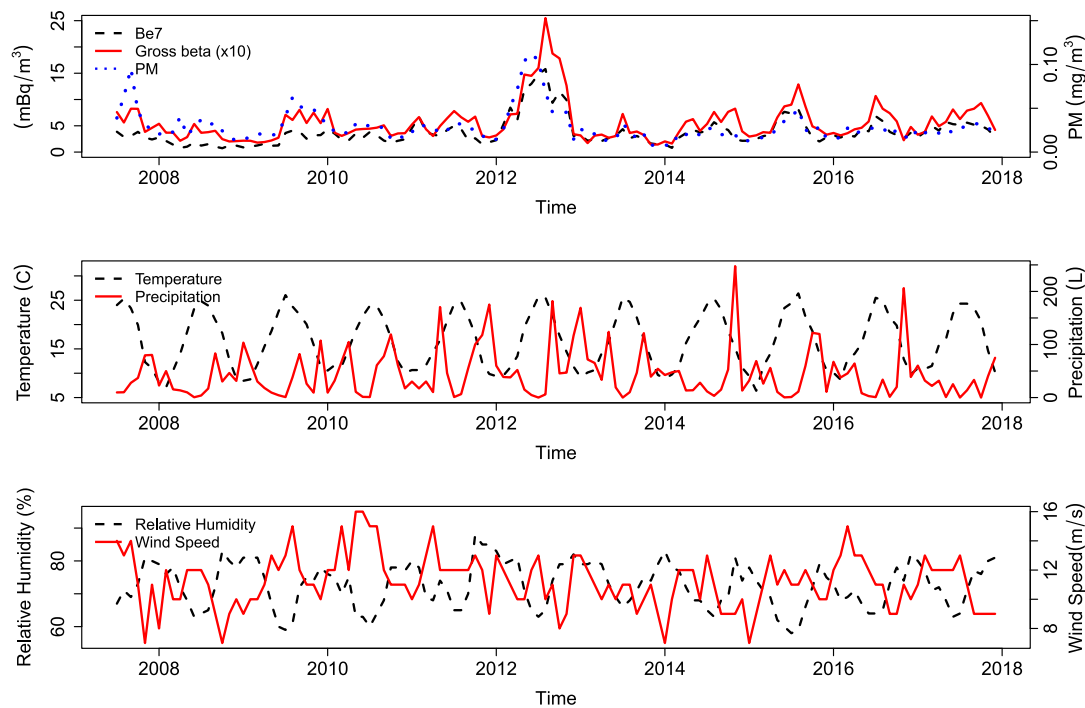


Fig. 2. Time series for all variables considered in this study and obtained between 2005 and 2014 at the University of the Balearic Islands. The values for the gross beta activity concentration have been multiplied by 10 to obtain a better visualization.

Table 1

Main descriptive statistical parameters for all physical magnitudes considered in this study.

Magnitude	Max.	Min.	Mean	standard deviation
PM[mg/m ³]	0.112	0.007	0.029	0.018
⁷ Be[mBq/m ³]	15.8	0.7	3.8	2.7
Gross beta[mBq/m ³]	2.55	0.14	0.56	0.37
Temperature [°C]	26.4	6.4	16.7	5.7
Precipitation[L]	248	0.00	45	47
Relative Humidity[%]	88	58	73	7
Wind speed[m/s]	16.0	7.0	11.2	1.8

previously been observed in other close locations like Valencia (Sáez-Muñoz et al., 2018), Granada (Piñero-García et al., 2012), Bilbao (Alegria et al., 2020), Malaga (Dueñas et al., 2009; Gordo et al., 2015) or Monaco (Pham et al., 2011), although in other distanced locations with different climate and atmospheric circulation it can be the opposite (Kitto et al., 2005). The 3.8 mBq/m³ average activity concentration for

⁷Be is comparable to other average values in Europe typically in the range of 2.5 and 6 mBq/m³ (Gordo et al., 2015).

Among all variables PM, ⁷Be, A_{β} and precipitation values follow a log-normal probability distribution, and a logarithm is applied to all of them to ensure that all data follow a normal distribution. Afterward, the pairwise correlation is quantified using Pearson's linear correlation coefficient. Note that this coefficient can be applied because we have transformed the data when necessary to ensure their normality. Alternatively, the Spearman correlation coefficient can be obtained for the non transformed variables with almost identical results. Almost all variables are correlated with high significance although, as observed in Table 2, the strength of their correlation can be quite different. A first group of highly positively correlated parameters (PM, ⁷Be, A_{β} and temperature) is immediately identified. The correlation between ⁷Be and A_{β} ($r = 0.85$, $p < 10^{-4}$) stands out in comparison to the rest. Dust concentration and temperature are also positively correlated to the activity concentrations. Relative humidity and precipitation are negatively correlated to ⁷Be and A_{β} , although these correlations are weaker. Wind speed is not significantly correlated neither to ⁷Be nor A_{β} activity

Table 2

Pearson's correlation coefficient between all variables on the data set including the seasonal component. The corresponding p-values are provided inside parenthesis.

	⁷ Be	Gross Beta	Temp	Prec	RH	WS
PM	0.59 ($< 10^{-4}$)	0.73 ($< 10^{-4}$)	0.52 ($< 10^{-4}$)	-0.23 (0.008 3)	-0.36 ($< 10^{-4}$)	0.25 (0.004 4)
⁷ Be		0.85 ($< 10^{-4}$)	0.38 ($< 10^{-4}$)	-0.21 (0.015 9)	-0.35 ($< 10^{-4}$)	0.15 (0.100 6)
Gross Beta			0.56 ($< 10^{-4}$)	-0.19 (0.030 1)	-0.36 ($< 10^{-4}$)	0.06 (0.537 8)
Temp				-0.33 (0.000 1)	-0.72 ($< 10^{-4}$)	0.24 (0.006 2)
Prec					0.50 ($< 10^{-4}$)	-0.14 (0.113 2)
RH						-0.50 ($< 10^{-4}$)

concentrations and does not seem to have an important direct effect on these magnitudes, although it is significantly correlated to the other meteorological variables.

The correlations found between the parameters characterizing the data set are actually expected considering their annual seasonal pattern. Still, it is relevant to compute them to compare with other studies where similar results are actually obtained and to check the accepted explanations of the temporal variation of ^7Be in the atmosphere. Both temperature and relative humidity have similar correlation in magnitude with ^7Be although of different sign, similarly to what it is found in Malaga (Dueñas et al., 2009) and Granada (Piñero-García et al., 2012). In both locations precipitation has a lower anticorrelation with ^7Be activity concentration, quite similarly to the case considered in this work. These are the most relevant meteorological parameters to explain the ^7Be activity concentration in the lower troposphere. High temperatures cause thermal convections, which enhances vertical exchange in the troposphere allowing ^7Be from the upper troposphere to reach the surface. Both precipitation and high relative humidity are connected to scavenging which reduces aerosol residence times in the atmosphere. While this allows to understand the common traits for the activity concentration in the atmosphere, it might be difficult to perform a detailed analysis where more direct connections between the meteorological parameters and the activity concentration values can be identified.

The seasonal components can strongly dominate these correlations and hide relations between the relevant parameters which might only become apparent in smaller subsets from the original data set. To overcome this problem it is possible to eliminate this seasonal dependence from the data set. All considered variables in this study can be considered as additive time series composed of a seasonal component which periodically repeats itself every year, a trend component which might smoothly change in time between different periods of time, and a remaining residue which can be interpreted as noise that can change fast for consecutive data points. The trend component can be obtained performing a moving average over one seasonal period (one year) window. This trend is then subtracted from the original time series and the

seasonal component is found averaging, for each month, over the entire time series. This seasonal component is centered afterward, to ensure that it has zero average. The effect of the seasonal component can then be eliminated subtracting the seasonal component from the original time series. The obtained time series represents the deviation from the seasonal average and, for this reason, it can take negative values.

Applying this process to all the variables, results in a new data set with quite different properties. A_β , ^7Be , and PM still change relatively smoothly and similarly over time, but their maximum and minimum values do not coincide necessarily any more with summer and winter periods. For the winters of 2008, 2010, and 2011 it is possible to observe in Fig. 3 that the three parameters reach local maxima, which is just the opposite behavior observed in the original data set. The temperature time series has a noisier pattern, although it is still possible to observe local maxima and minima which do no longer always coincide with the maxima and minima of the activity concentrations. While in the winter of 2011 the temperature also reaches a local maxima, during the summer of 2012 when the absolute maximum for A_β , ^7Be , and PM is reached, the temperature has average values. Precipitation, relative humidity and wind speed resemble a noisy time series with fast alternation between high and low values, and their time variation seems to be decoupled from the rest of the parameters. The correlations between the variables of this new data set, quantified with their corresponding correlation coefficient in Table 3, are quite different to those obtained in the original data set and match the behavior observed in the time series plots. The set of highly significantly correlated parameters (A_β , ^7Be , PM, and Temp) is still kept, although now the correlations with the temperature are weaker. The remaining correlations are lost, which means that the source of these correlations in the entire original data set can be almost entirely explained by its seasonal component. Only the relative humidity retains a marginal correlation with precipitation and wind speed.

All correlation coefficients and statistical parameters as well as the data time series treatment have been implemented using the R language for statistical computing (R Core Team, 2020).

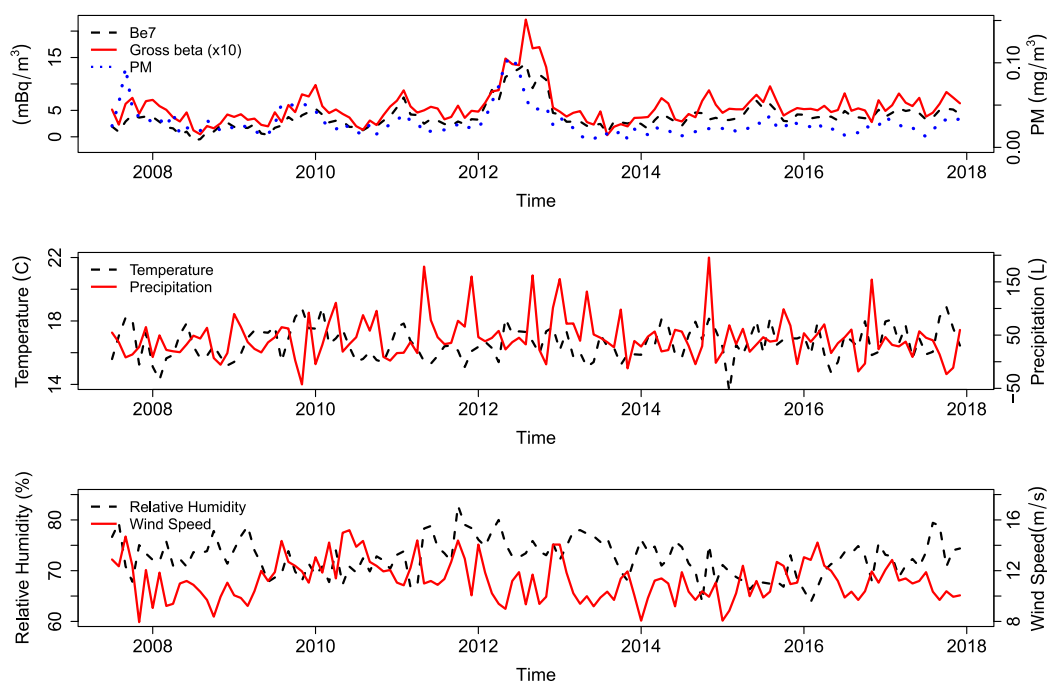


Fig. 3. Time series without seasonal component for all variables considered in this study and obtained between 2005 and 2014 at the University of the Balearic Islands. The values for the gross beta activity concentration have been multiplied by 10 to obtain a better visualization. The values represent variations from the seasonal average and for this reason in some cases negative values are obtained.

Table 3

Pearson's correlation coefficient between all variables on the data set from which the seasonal component has been subtracted. The corresponding p-values are provided inside parenthesis.

	^7Be	Gross Beta	Temp	Prec	RH	WS
PM	0.55 ($< 10^{-4}$)	0.59 ($< 10^{-4}$)	0.34 (0.000 1)	-0.11 (0.200 0)	-0.06 (0.520 8)	0.15 (0.102 4)
^7Be		0.86 ($< 10^{-4}$)	0.28 (0.001 6)	-0.10 (0.271 8)	-0.04 (0.677 3)	-0.12 (0.176 1)
Gross Beta			0.35 ($< 10^{-4}$)	-0.06 (0.472 0)	-0.06 (0.490 6)	-0.07 (0.422 8)
Temp				-0.27 (0.002 2)	-0.22 (0.014 5)	0.12 (0.167 8)
Prec					0.25 (0.005 2)	0.14 (0.114 7)
RH						-0.33 (0.000 1)

2.5. Dimension reduction techniques

Data analysis is highly simplified with the aid of the visual inspection of the data to unravel relations between different points and the variables that characterize them, as well as to identify families and clusters of similar points. All data contained in a data set with N variables can be seen as points embedded in a N-dimensional space, where its coordinates are the corresponding N variable values for each data. In this context, the degree of similarity between two data points can be quantified by the distance between them, such that close points tend to represent data with similar properties. The choice of a particular distance in this space defines how these data points are distributed and actually unravels different relations among them, as well as the characterization of the whole data set. The most common distance to be used is actually the Euclidean distance.

$$d_{\text{Eucl}}(x, y) = \sqrt{\sum_{i=1}^N (x_i - y_i)^2}, \quad (1)$$

where x_i and y_i represent the values for each of the N variables of the data set for the x and y data points.

These graphical representations are hard to comprehend for data sets characterized with a large number of variables, as they are embedded in a high dimensional space. Dimension reduction techniques map these high dimensional structures into a lower dimensional space, usually with two or three dimensions, where data can easily be represented and visualized.

The goal of these techniques is to preserve as much of the characteristic structure in the high dimensional space and map it into the lower dimensional one, with the minimal possible loss of information. Some of these algorithms focus on keeping close in the low dimensional map those points which are similar in the original space, it is then said that the local structure is preserved. A different option is to keep original dissimilar points as far away as possible in the lower dimensional space, preserving the global structure of the data set which can potentially distort the local structure in the new transformed space. The selection of a particular dimension reduction technique is frequently a prioritization of either the global or local structure of the data set. Focusing on a local structure preservation can help identifying clusters of points with similar properties, facilitating classification tasks, although the loss of global structure means that the distances between clusters should not necessarily be interpreted as a measure of the magnitude of the differences between the characteristics of these clusters. The alternative approach, preserving the global structure, maintains dissimilar points further away and enables the interpretation of the global effect of each variable, but commonly implies losing local information and makes it difficult to identify clusters of similar data points.

All reduction algorithms applied in this work (PCA, t-SNE and UMAP), which will be explained in detail in the following subsections,

have been implemented using Python programming language. For PCA and t-SNE the corresponding functions in the scikit-learn library (Pedregosa et al., 2011) have been used, while the umap library has been used for the remaining algorithm (McInnes et al., 2018).

2.5.1. Principal Component Analysis (PCA)

PCA is a technique applied to data sets characterized by a series of correlated variables. It assumes linear relationships among the original variables and applies on them a linear combination which provides a new set of uncorrelated variables known as Principal Components (PC). These PCs are sorted by the amount of variance of the original data set contained in each PC. This property is especially useful to visualize the data set in a new lower dimension space without a significant loss of information. Its main drawback is that this transformation is designed to keep as much information of the global structures as possible, in detriment of the local structure of the data set, which determines the formation and identification of clusters with different properties. When the variables in the original data set correspond to different magnitudes it is advisable to standardize them, such that all variables are centered around 0 and have unit standard deviation.

2.5.2. t-distributed Stochastic Neighbor Embedding (t-SNE)

t-SNE is a non-linear dimensionality reduction technique aimed at mapping a data set in a high dimensional space to a new one with 2 or 3 dimensions, while keeping and revealing the underlying structure at different scales. It is a variation of the Stochastic Neighbor Embedding (SNE) (Hinton and Roweis, 2002) which simplifies the optimization process and improves the visualization in the new low dimensional space (van der Maaten and Hinton, 2008).

In t-SNE the distances between points of the data set are converted into symmetrical probabilities between points in such a way that similar points are assigned a higher probability. The probability distribution in the high dimensional space is constructed using a Gaussian distribution, while the probability distribution in the lower dimensional space is obtained using a t-Student distribution with 1 degree of freedom. This last choice is one of the main differences to the SNE algorithm and increases the distance between close clusters of data, as well as simplifies the optimization process. Finally, the Kullback-Leibler divergence, a measure of dissimilarity between probability distributions, is minimized in respect of the positions of the data points in the new lower dimensional space. A more complete description of the algorithm is explained in the detail in (van der Maaten and Hinton, 2008).

The main parameter of this technique is the so-called perplexity, which represents the effective number of neighbors of a given point. Typical values range between 5 and 50, and its value must be tuned in each case depending on whether more importance is given to the local or to the global structure.

2.5.3. Uniform Manifold Approximation (UMAP)

UMAP is a recently introduced manifold learning technique which provides significant advantages to perform dimension reduction (McInnes et al., 2018, 2020). The visualizations obtained using UMAP are comparable to those obtained with t-SNE, although the former keeps a better global structure. Additionally, UMAP is characterized by a set of parameters which improve the quality of the final result, scales better with the dimension of the data set, and is faster than t-SNE.

The topology on the original and new spaces is characterized defining a finite neighborhood around each point and counting the number of intersections between them. This process is performed applying advanced mathematical tools from fuzzy topology. In fuzzy set theory, the membership to a given set is not binary (yes or no) but it actually has several degrees of membership. The lower dimensional space is found through an iterative optimization process ensuring that the fuzzy topological representation in the new space is as close as possible to the original one. A more detailed explanation of UMAP's mathematical basis can be found in (McInnes et al., 2020).

Applying UMAP requires tuning several parameters: the number of neighbors considered for each point, which allows to balance the importance of local versus global structure; and the minimum distance between points in the low dimensional space, which sets how packed the clusters will be. Several alternatives can be used to compute the distance between points, being the most frequently used the Euclidean (Eq. (1)), cosine, and correlation distances. The cosine distance is related to the cosine similarity which provides the cosine of the angle between two non-zero vectors of an inner product space. As opposed to the Euclidean distance, which mainly depends on the absolute value of each variable, it provides a relative weight to each variable in a given vector.

$$d_{cos} = 1 - \text{cosine similarity}; \quad (2)$$

$$\text{cosine similarity} = \frac{\sum_{i=1}^N (x_i y_i)}{\sqrt{\sum_{i=1}^N x_i^2} \sqrt{\sum_{i=1}^N y_i^2}} \quad (3)$$

The correlation distance is obtained from Pearson's linear

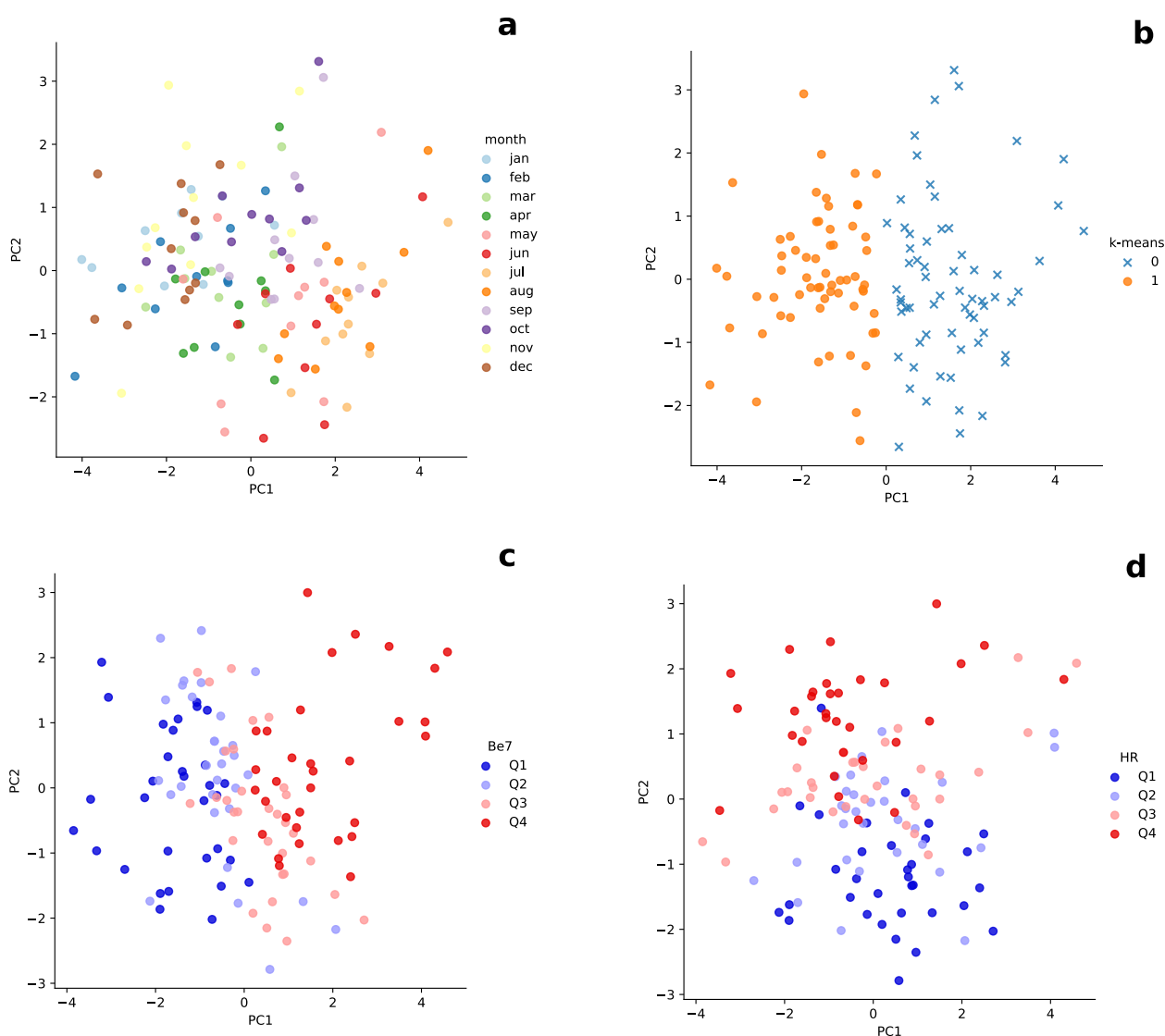


Fig. 4. Representation of all measurements in the two dimensional space obtained after applying the PCA algorithm to the original dataset (sub-figures a and b). The first Principal Component (PC1) explains 50.0% of the original variance, while the second Principal Component (PC1) explains 19.1%. The measurements are represented according to their corresponding month of the year in sub-figure a, while their ascription to one of the two clusters obtained after applying k-means clustering is represented in sub-figure b. In the lower row (sub-figures c and d), two PCA representation for the dataset without the seasonal component are plotted, the first Principal Component (PC1) explains 37.2% of the original variance, while the second Principal Component (PC1) explains 20.5%. Sub-figure c corresponds to different ^{7}Be activity concentration values while sub-figure d represents the role of relative humidity. In both cases the points are divided in 4 groups corresponding to their corresponding quartiles for each variable.

correlation coefficient (r), and it is zero for perfect correlation:

$$d_{\text{Corr}}(x, y) = 1 - r; \quad (4)$$

$$r = \frac{\sum_{i=1}^N (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^N (x_i - \bar{x})^2} \sqrt{\sum_{i=1}^N (y_i - \bar{y})^2}} \quad (5)$$

Additionally, and in opposition to how t-SNE works, once the embedding in the lower dimensional space has been obtained it is possible to add new points to it, without having to repeat the entire optimization process.

3. Results and discussion

3.1. Principal Component Analysis

As a first step, the well known and commonly used dimension reduction technique in these kind of studies, Principal Component Analysis (PCA) is applied to the data set described in section 2. As a consequence of the PCA tendency to maintain dissimilar points far away from each other, the majority of points are packed in the central region of the new space, as can be seen in Fig. 4. Given that all variables follow a similar seasonal pattern and are then correlated, all points are labeled according to their corresponding months to provide indication of the typical values for all variables. While there is not a clear cluster pattern formation, it is possible to observe a trend of points corresponding to hotter and drier months to lie on one side of the space along PC1, while the rest of the points lie on the opposite side.

In order to identify clusters in such space, the k-means clustering algorithm is applied to this data set. Two different clusters appear, one for PC1 positive values which can be identified with hotter months, and a second one corresponding to the coldest months located on the negative values PC1 region. It is not a surprise that the data set can be explained with a reduced number of large and not distanced clusters, given the importance of the season components. This visualization mainly reflects the western Mediterranean climatological properties of the sampling site.

Applying the PCA to the second data set, where the seasonal component has been subtracted, provides a similar representation in sub-figures c and d. Without the seasonal component, the month of each point does not provide any relevant information so all data points are labeled according to their inclusion in one of the four quartiles for the corresponding variable. The lowest 25% of values are labeled as Q1, the following 25% as Q2, and so on. Quite similarly to what happens with the original data set, high ^7Be activity concentration values lie in the positive PC1 region, while the lowest values are identified with negative PC1 values. The distribution according to the gross beta activity and dust concentrations is almost identical, while using the temperature to label the points results in a similar although blurrier representation. The second component PC2 mainly encodes the information related to the relative humidity, with the highest values identified with positive PC2 values. Wind speed and precipitation are not so clearly distributed in this new lower dimensional space. This classification is actually expected as PCA provides uncorrelated new variables keeping as much global information as possible. It is only natural that ^7Be , A_β , PM and temperature are aligned along the same PC as they share common and repeated information, while the second PC is left for other uncorrelated variables. Also, the global information preservation provides a representation which enables to interpret the role of the original variables in the new space, which is one of the main strengths of PCA, but the process is applied filtering away key information that causes mixing of clusters of similar data points. For this reason PCA is a good tool to interpret the global conditions of a data set, but lacks capacity to implement classifications tasks.

There is no clear way to decide how many clusters should be set for both data sets. Also, and given the global structure of the data in this

space, the boundary between the clusters would never be clear. Cluster identification in the new space obtained applying PCA is then neither reliable nor robust against the clustering algorithm tuning. This is a known PCA property for these kind of data sets where the differences between potentially existing clusters are actually small and sharp boundaries between them are not to be expected. However, this faithful representation of the global properties of the data set is actually one of PCA main advantages as it helps to obtain an interpretation and explanation of the entire data set.

3.2. t-distributed stochastic neighbor embedding (t-SNE)

Low dimensional spaces obtained using t-SNE prioritize local structure preservation. Perplexity, which can be connected to the number of neighbors considered for each point, is the algorithm parameter with the highest impact on the final result. Setting perplexity equal to 5, the very local structure of the data set emerges in subfigure 5a revealing a large number of small clusters, although the separation between these clusters is not clear. Increasing the perplexity value shifts the focus to a more global structure, where the clusters are not so packed and tend to mix, as can be seen in sub-Fig. 5b and 5c, corresponding to perplexity values of 10 and 20 respectively.

For the lowest perplexity value (perplexity = 5) it is indeed possible to identify well defined small clusters, which include points corresponding to close months, sharing similar properties. Summer month points, with high ^7Be and A_β values, lie in the negative value region of dim1 and dim2, while the remaining clusters are not so easy to identify. There are some small clusters with similar points, but their structure is not clear and it is possible to find clusters which in principle are quite similar but nonetheless are quite far away in this new space. In the other extreme, the structure obtained in Fig. 5c using a high perplexity value (20) does not provide a clear indication of well defined clusters. It actually resembles the visualization obtained with PCA in Fig. 5, but without the proper preservation of the global structure neither the repeatability offered by PCA.

The original global structure of the data set is not well preserved by t-SNE so, while close points in the new space are in principle also close in the original space, the same does not necessarily hold for distant points. Running several successive times the t-SNE on the same data set actually produces different outcomes. Although the composition of these clusters is almost always the same, their location and the relative separation among them can indeed change.

Moving to the data set without the seasonal components provides a t-SNE representation slightly easier to interpret. The chosen example in Fig. 5 corresponds to a perplexity value equal to 5. Using a k-means clustering it is possible to identify 3 different clusters in subfigure 5d. Cluster 0 is characterized by low ^7Be , A_β and PM values, and high relative humidity values. Cluster 1 has high values of gross beta, ^7Be dust content, while cluster 2 can be identified with low values of relative humidity. The rest of variables has a mixed classification among the three clusters, although the temperature has a somewhat similar behavior to ^7Be . Comparing this representation to the PCA one, it is easier to identify and locate different clusters, although the highest and lowest values for each variable are globally better separated in the PCA picture. This is a known drawback of t-SNE as it does not preserve the global structure of the data set which, combined with the low repeatability of t-SNE, doesn't make this algorithm recommendable for this particular data set.

3.3. Uniform Manifold Approximation and Projection (UMAP)

Better data set visualizations and cluster identification are represented in Fig. 6 applying UMAP using the Euclidean, cosine, and correlation metrics and considering different number of neighbors for each point. The best balance between local and global structure for the original data set is actually obtained setting the number of nearest neighbors to 10. One can then compare in Fig. 6a, b, and 6c the results

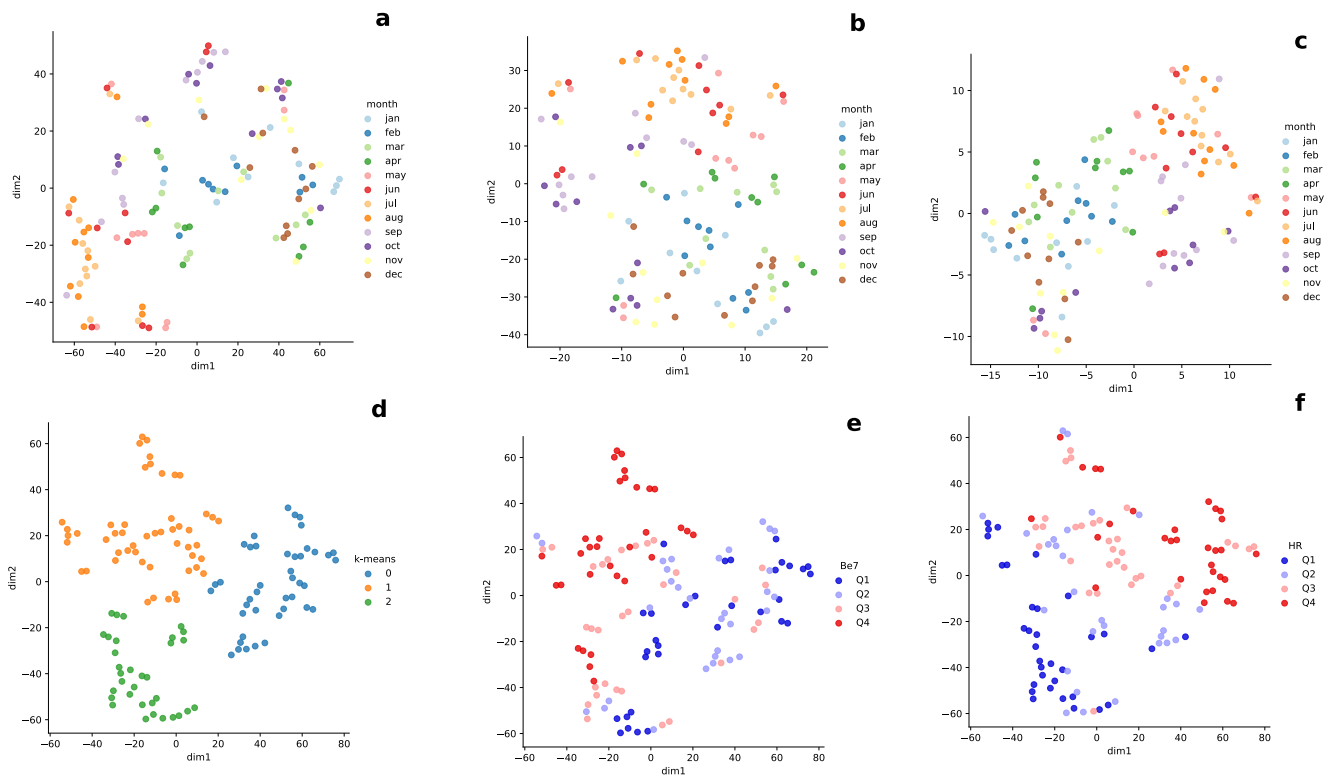


Fig. 5. t-SNE representation for the original data set with different perplexity values in sub-figures a, b and c. The measurements are represented according to their corresponding month of the year in these sub-figures. In sub-figure a the local structure prevails using perplexity = 5, while in sub-figure b (perplexity = 10) all clusters become closer. In sub-figure c (perplexity = 20) the global structure without clear cluster formation is the dominant visualization. The last three sub-figures (d, e and f) represent the t-SNE transformation for perplexity = 5. The three clusters obtained applying the k-means algorithm are plotted in sub-figure d. Sub-figure e corresponds to different ${}^7\text{Be}$ activity concentration values while sub-figure f represents the role of relative humidity. In both cases the points are divided in 4 groups corresponding to their corresponding quartiles for each variable.

obtained using the correlation, Euclidean, and cosine metrics respectively. While the correlation distance provides two well defined clusters, using the Euclidean and cosine distances several not so well defined small clusters appear.

The Euclidean metric separates the points corresponding to warmer months from those corresponding to colder points, along the dim1 axis. Increasing the number of neighbors per point the global structure prevails, mixing the clusters that become evident for lower number of neighbors. In this respect, performing UMAP using the Euclidean distance in this data set provides similar results to the t-SNE visualization previously obtained.

The cosine distance does also visually reveal the existence of several clusters. In this case, this clustered structure does prevail with the increasing number of considered neighbors, although both the number of clusters and the identification of points to a given cluster change with these parameters. The different obtained visualizations might provide different insights, although it is in principle difficult to justify the election of one in front of the other in this particular case.

On the contrary, the correlation distance does not only provide the visualization with the best cluster identification, but the structure of those clusters is robust under changes in the number of considered neighbors, as it is the assignation of each point to either cluster.

The UMAP algorithm using the correlation distance does then provide a useful tool to identify the existence of two well defined clusters in this data set. To confirm these findings, a k-means clustering algorithm is applied in the UMAP generated 2-dimensional space. The obtained clusters are represented in subfigures 6d, 6e and 6f for increasing number of neighbors. In all cases the obtained clusters match those that were already visible in Fig. 6a. Cluster 0 corresponds to the cluster that appears for positive dim1 values in Fig. 6a, while the points in the negative dim1 region correspond to those belonging to cluster 1. The

changing cluster orientation and location observed in these figures is common for different UMAP realizations and has no further physical implications as both the internal structure within each cluster and the distances between the two clusters remain fairly the same. To check the robustness of cluster identification, the HDBSCAN clustering algorithm is applied to the same spaces, yielding the same results obtained by k-means. The number of clusters to be identified is not established beforehand in the HDBSCAN algorithm, and points can be assigned no cluster and be identified as noise. These results show then the usefulness of UMAP using the correlation distance to clearly identify groups of measurements with common properties.

This first cluster, identified as cluster 0 in Fig. 6 is composed of 68 points, and corresponds to measurements belonging to warm and dry months, as can be seen in Table 4. These points are characterized by high values of ${}^7\text{Be}$ and A_β concentration of activity, with mean values of 4.7 mBq/m^3 and 0.71 mBq/m^3 . Contrary, points belonging to cluster 1 are identified with colder and wetter months, characterized by lower values of ${}^7\text{Be}$ and A_β concentration of activity with mean values of 2.7 mBq/m^3 and 0.39 mBq/m^3 (see Table 5). UMAP is then able to clearly uncover the existence of two different clusters with higher and lower activity concentration, each of them characterized with their own local climatological properties. A similar relation between ${}^7\text{Be}$ and local meteorological properties had also been found previously (Piñero-García and Ferro-García, 2013), although in those cases PCA was used to obtain the lower dimensionality visualization, and the identification of clusters was not so clear.

The analysis of the UMAP representation of the data set without the seasonal components provides a quite different scenario to all the previous ones. A finer cluster structure is revealed in Fig. 7, with a higher number of clusters with identifying properties. A k-means clustering is applied to label 8 different clusters, with the aim to identify the

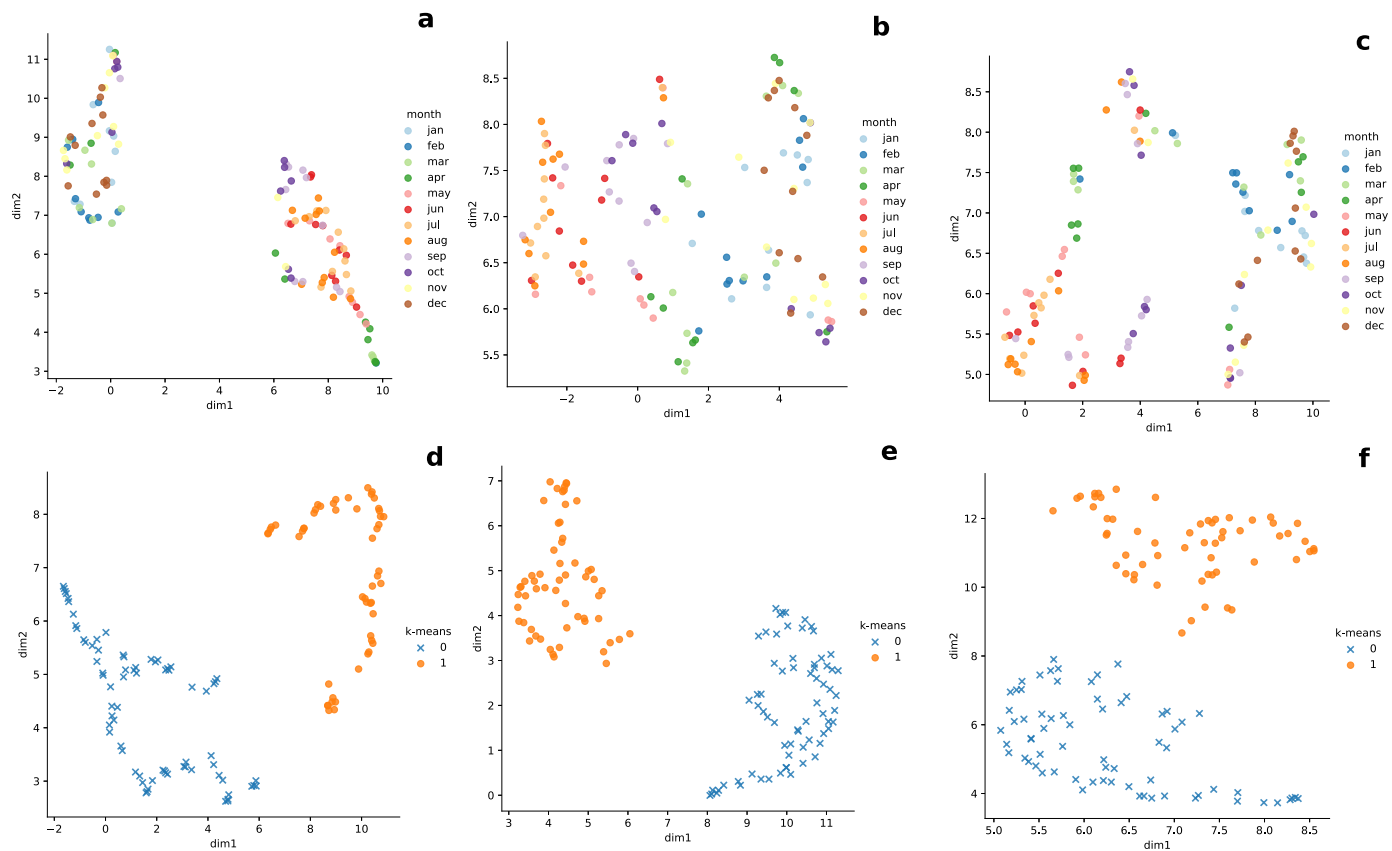


Fig. 6. UMAP representation of the data set. In the first row (sub-figures a, b and c) 10 neighbors are considered for each point, and all measurements are represented according to their corresponding month of the year. Each sub-figure from the first row is obtained using a different distance: correlation (a), euclidean (b) and cosine (c). Figures in the second round are all obtained using the correlation distance for different number of neighbor points: 5 (d), 15 (e) and 20 (f) where each point is labeled according to the k-means cluster to which it belongs. All clusters obtained with the correlation distance are equivalent up to a global rotation and minor differences in their internal structure.

Table 4

Main descriptive statistical parameters for the 68 points included in the first cluster obtained by k-means clustering in the set-up represented in Fig. 6.

Magnitude	Max.	Min.	Mean	standard deviation
PM[mg/m ³]	0.112	0.014	0.036	0.021
⁷ Be[mBq/m ³]	15.8	0.8	4.7	3.1
Gross beta[mBq/m ³]	2.55	0.22	0.71	0.42
Temperature [°C]	26.4	8.7	20.1	4.2
Precipitation[L]	182	0	24	29
Relative Humidity[%]	79	58	68	5
Wind speed[m/s]	16.0	8.0	11.8	1.8

Table 5

Main descriptive statistical parameters for the 58 points included in the second cluster obtained by k-means clustering in the set-up represented in Fig. 6.

Magnitude	Max.	Min.	Mean	standard deviation
PM[mg/m ³]	0.072	0.007	0.021	0.010
⁷ Be[mBq/m ³]	14.6	0.7	2.7	1.5
Gross beta[mBq/m ³]	0.83	0.14	0.39	0.17
Temperature [°C]	20.9	6.4	12.0	3.1
Precipitation[L]	248	9	58	69
Relative Humidity[%]	88	71	78	4
Wind speed[m/s]	13.0	7.0	10.5	1.6

differences among them. To facilitate the analysis of the common properties of each cluster, all data points are labeled according to their quartile classification according to all seven parameters of the data set.

High positive deviations of ⁷Be can be linked to clusters 1, 5 and 7,

although it is possible to find differences among these clusters related to the rest of variables.

Both clusters 5 and 7 have the expected high temperature and A_{β} , although cluster 5 can be additionally associated to negative precipitation deviations and cluster 7 to negative relative humidity deviations. These are situations where thermal convection processes that facilitate ⁷Be exchange in the troposphere are enhanced, although without apparent direct effect on deviations in respect to the seasonal average for A_{β} and dust content. However, in cluster 1 high relative humidity is combined with low temperature. These high ⁷Be and A_{β} combined with high relative humidity in cluster 7 is not that common, because high HR is usually associated to higher size aerosol formation and lower residence time in the atmosphere.

High negative deviations of ⁷Be characterize clusters 3, 4 and 6. All these clusters are also endowed with high relative humidity values. It is possible to identify two different groups in cluster 4, with opposite values A_{β} and PM, although both cases are combined with high precipitations. High negative deviations for A_{β} , PM and wind speed are combined with high negative deviations of ⁷Be in cluster 5.

Finally, the remaining clusters 0 and 2 are not clearly connected to ⁷Be and A_{β} . Low relative humidity and high winds characterize cluster 0, while cluster 3 can be clearly connected to low temperatures.

It is worth noting that in this second data set the HDBSCAN clustering algorithm has also been applied obtaining an almost identical cluster identification with 8 different clusters. The only difference is that HDBSCAN joins clusters 1 and 2 in a single cluster, and split cluster 4 in two different clusters. This alternative classification gives more importance to the similarities between ⁷Be and A_{β} , although it is not so easier to identify the role of temperature.

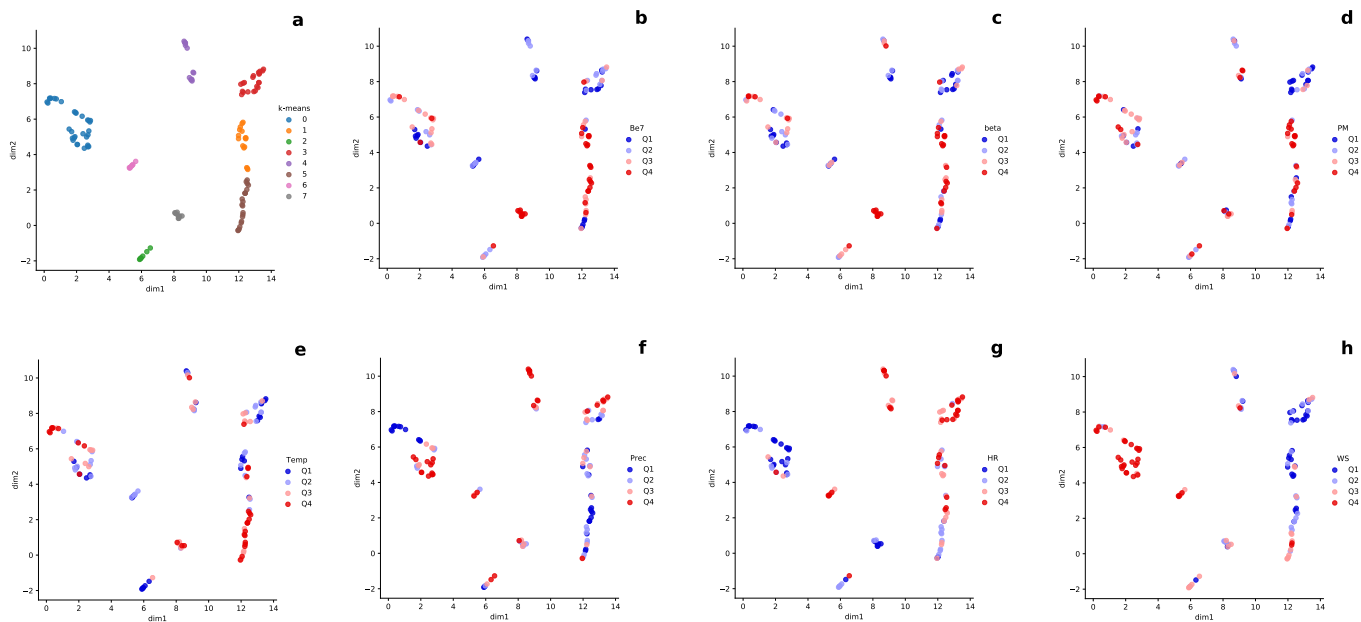


Fig. 7. UMAP representation for the deviations from seasonal average data set using the correlation distance and 5 neighbors per data point. The 8 clusters identified by a k-means clustering algorithm are plotted in sub-figure a. The seven data set variables are represented in the rest of sub-figures: ^7Be (b), gross beta (c), PM (d), temperature (e), precipitation (f), relative humidity (g) and wind speed (h).

In summary, with this representation it is possible to identify sub-clusters of states with different relations between ^7Be and the rest of variables deviations respect their corresponding seasonal typical values. High ^7Be values are usually combined with high temperature deviations, although it is still possible to achieve those values during months with low temperature and high relative humidity deviations. The absence of wind might explain an anomalous higher residence times for aerosols, which increase the typical activity concentration values. However, high negative ^7Be deviations are clearly linked to high relative humidity, which usually increases aerosol size and causes shorter aerosol residence times in the atmosphere. It is though possible to identify different clusters within this group, characterized by their own A_{β} , PM and wind deviations from the average. UMAP helps in this case to identify different groups of data points with particular relations between the activity concentrations and meteorological conditions which are not evident with alternative visualizations, obtained with PCA and t-SNE, and from the global correlations analysis.

4. Conclusions

The PCA technique has already been proven to be extremely useful in environmental radioactivity studies to visualize the existing relation between activity concentrations and meteorological parameters, but it has a poor capacity at identifying and visualizing clusters of measurements with similar characteristics. The t-SNE algorithm applied to the data set considered in this work, especially for low perplexity values, reveals finer structure although some information on the global structure is lost in the process. Although in this particular case it is not a great improvement compared to PCA, it already indicates its potential to reveal relevant information in larger data set with a richer local structure that might be hidden in PCA visualizations.

To obtain a useful visualization with the right balance between global and local structure, UMAP is the better option. While one has to find the correct set of algorithm parameters, it clearly provides a useful tool to visually identify the existence of clusters composed by measurements with similar properties. In this case, the use of the correlation

distance has proven to be the most useful to reveal the cluster structure in the obtained low dimensional spaces. In the original data set, where the seasonal components dominate, UMAP clearly identifies two well defined clusters separating higher activity concentration measurements from those with lower activity values, where each cluster has also different characteristic meteorological properties. This identification is robust under UMAP's parameter tuning and the choice of the clustering algorithm. When the seasonal component is subtracted from the data set, a finer structure with subtler relations between the variables appears, enabling the identification of scenarios which are not clear neither from the global correlation analysis nor from the visualizations obtained with alternative dimension reduction techniques. UMAP proves then to be useful both when the global and local structure dominate the data set characterization and interpretation.

The choice of the dimension reduction technique to apply in a particular case highly depends on the information that is wanted to gather. PCA provides a faithful representation of the global properties of the data set and the dimensions on the low dimensional space are clearly linked to the dimensions (variables) in the initial space providing an easy interpretation, this is one of its main advantages. On the contrary, t-SNE and UMAP are designed to magnify the differences between clusters of similar data points. The dimensions of the transformed spaces are not clearly connected to the original variables, which is the main drawback of these techniques, but it is easier to identify clusters of similar states which is of great help for classification tasks. UMAP in particular provides a useful balance between identification of clusters and preservation of the global data set information. UMAP has already proved to be quite successful in other fields, and is a promising technique to gather useful insights in environmental radioactivity studies, especially when a large number of points and variables are considered.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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